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職歴

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学歴

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2012/4 – 2018/3	東京大学　医学部医学科

受賞歴

2026/2/4	井上科学振興財団　井上研究奨励賞
2025/12/19	日本人類遺伝学会　第 70 回大会 大会最優秀口演賞
2024/10/10	日本人類遺伝学会　2024 年ヨーロッパ人類遺伝学会 Annual Meeting トラベルアワード
2024/3/31	大阪大学大学院医学系研究科　博士課程優秀者

競争的資金

2026/4 – 2028/3	日本学術振興会　科学研究費助成事業 若手研究「深層学習によるヒト脂質代謝の遺伝的基盤全体像の解明」（研究代表者）
2025/10 – 2028/3	日本医療研究開発機構（AMED）　ゲノム創薬基盤推進研究事業「遺伝子変異の機能性スペクトラム統一的解析と次世代創薬連携による創薬シーズ導出」（研究代表者）
2024/7 – 2027/3	日本医療研究開発機構（AMED）　ゲノム医療実現推進プラットフォーム・先端ゲノム研究開発 (GRIFIN)「遺伝子–環境相互作用の学術・オミクス横断による個別化医療の実装」（研究代表者）

2024/6 –	日本応用酵素協会　Cardiovascular Innovative Conference に関する研究助成 (CVIC)「心血管疾患の遺伝子–環境相互作用解明によるゲノム個別化医療と創薬」（研究代表者）

奨学助成

2020/4 – 2023/9	武田科学振興財団　医学部博士課程奨学助成

教育歴

2026/4 –	医学に接する　東京大学　医学部
2025/4 –	生化学講義　東京大学　医学部
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主要学術論文

* 共同筆頭著者、♯ 責任著者（共同責任著者を含む）・査読付き論文 40 編のうち主要な 6 編。全リストは本 CV 末尾に記載。

1. **Namba S♯**, Sonehara K, Koyanagi YN, Kikuchi T, Ojima T, Edahiro R *et al.* A cross-population compendium of gene–environment interactions. *Nature* 688–697 (2026).
2. Caro I*, Western D*, **Namba S***, Sun N, Kawaguchi S, He Y *et al.* Proteogenomics in cerebrospinal fluid and plasma reveals new biological fingerprint of cerebral small vessel disease. *Nature Aging* 2514–2531 (2025).
3. **Namba S♯**, Akiyama M, Hamanoue H, Kato K, Kawashima M, Kushima I *et al.* Inconsistent embryo selection across polygenic score methods. *Nature Human Behaviour* 2264–2267 (2024).
4. **Namba S***, Saito Y*, Kogure Y, Masuda T, Bondy ML, Gharahkhani P *et al.* Common germline risk variants impact somatic alterations and clinical features across cancers. *Cancer Research* **83,** 20–27 (2022).
5. **Namba S***, Konuma T*, Wu KH, Zhou W & Okada Y. A practical guideline of genomics-driven drug discovery in the era of global biobank meta-analysis. *Cell Genomics* **2,** 100190 (2022).
6. Mishra A*, Malik R*, Hachiya T*, Jürgenson T*, **Namba S***, Posner DC* *et al.* Stroke genetics informs drug discovery and risk prediction across ancestries. *Nature* **611,** 115–123 (2022).

研究分野

遺伝統計学 / 集団遺伝学 / がん / ゲノムワイド関連解析 / 選択圧 / ゲノム情報にもとづいた疾患予測・薬剤開発

研究概要

ヒトゲノムを対象とした遺伝統計学研究に従事している。バイオバンク規模のデータを用いて、複合形質の遺伝的構造の解明、遺伝子–環境相互作用の解析、多遺伝子リスクスコアによる疾患予測、およびゲノム情報にもとづく創薬標的の同定を主要な研究課題としている。

公開プロフィール

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学術論文

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プレプリント

1. Palmer DS, Hill B, Hodgson S, Jõeloo M, Kalantzis G, ..., **Namba S** *et al.* The Biobank Rare Variant consortium powers the discovery of rare genetic associations through global collaboration. *medRxiv* (2026).
2. Kyosaka T, Narita A, Kulski JK, Minn AKK, Miyake A, ..., **Namba S** *et al.* Genome-wide association and Mendelian randomization analyses link *Helicobacter pylori* infection to Human Leukocyte Antigen polymorphisms and autoimmune diseases. *medRxiv* (2026).
3. Verma S, Guare L, Das J, Caruth L, Rajagopalan A, ..., **Namba S** *et al.* Expanding the genetic landscape of endometriosis: Integrative -omics analyses implicate key genes and pathways in a multi-ancestry study of over one million women. *Res Sq* (2025).
4. Yang C, **Namba S**, Matsuda K, Okada Y, Moran L, Vincent A & Marques FZ. Acetate, a fibre-derived gut metabolite, and modification of hormone-related cardiovascular risk in females. *medRxiv* (2026).
5. Uffelmann E, Wightman DP, Bahrami S, Shadrin AA, Fominykh V, ..., **Namba S** *et al.* Genomic analyses reveal new insights into Alzheimer’s disease. *medRxiv* (2025).
6. Shiraishi Y, Ochi Y, Sugawa M, Sakamoto Y, Kimura K, ..., **Namba S** *et al.* Rare k-mers reveal centromere haplogroups underlying human diversity and cancer translocations. *bioRxiv* (2025).

査読付き学術論文

1. Sato G, Yamamoto Y, Sonehara K, Saiki R, Ojima T, ..., **Namba S** *et al.* Genetic regulation across germline and somatic variation on the Y chromosome contributes to type 2 diabetes. *Nature Medicine* 894–905 (2026).
2. Ishigami D*, **Namba S***, Miyawaki S, Mitsui J, Imai H, Shimizu M *et al.* Delineating the Genetic Basis of RNF213-related vasculopathies: The association of PKHD1 variants with bilateral cerebral vasculopathy. *European Journal of Human Genetics* 788–795 (2026).
3. White SL, Brasher MS, Pattee J, Zhou W, Chapman S, ..., **Namba S** *et al.* Global multi-ancestry genome-wide analyses identify genes and biological pathways associated with thyroid cancer and benign thyroid diseases. *Nature Genetics* **58,** 307–316 (2026).
4. **Namba S♯**, Sonehara K, Koyanagi YN, Kikuchi T, Ojima T, Edahiro R *et al.* A cross-population compendium of gene–environment interactions. *Nature* 688–697 (2026).
5. Caro I*, Western D*, **Namba S***, Sun N, Kawaguchi S, He Y *et al.* Proteogenomics in cerebrospinal fluid and plasma reveals new biological fingerprint of cerebral small vessel disease. *Nature Aging* 2514–2531 (2025).
6. Edahiro R, Sato G, Naito T, Shirai Y, Saiki R, ..., **Namba S** *et al.* Deciphering state-dependent immune features from multi-layer omics data at single-cell resolution. *Nature Genetics* **57,** 1905–1921 (2025).
7. Sonehara K, Watanabe R, Matsumura Y, Mitsui Y, Ogawa Y, ..., **Namba S** *et al.* Whole-genome sequencing reveals rare and structural variants contributing to psoriasis and identifies CERCAM as a risk gene. *Cell Genomics* **5,** 100978 (2025).
8. Yamamoto Y, Shirai Y, Sonehara K, **Namba S**, Ojima T, Yamamoto K *et al.* Dissecting cross-population polygenic heterogeneity across respiratory and cardiometabolic diseases. *Nature Communications* **16,** 3765 (2025).

9. Yata T, Sato G, Ogawa K, Naito T, Sonehara K, ..., **Namba S** *et al.* Contribution of germline and somatic mutations to risk of neuromyelitis optica spectrum disorder. *Cell Genomics* **5,** 100776 (2025).
10. Sonehara K, Uwamino Y, Saiki R, Takeshita M, **Namba S**, Uno S *et al.* Germline variants and mosaic chromosomal alterations affect COVID-19 vaccine immunogenicity. *Cell Genomics* **5,** 100783 (2025).
11. Sasa N, Kojima S, Koide R, Hasegawa T, Namkoong H, ..., **Namba S** *et al.* Blood DNA virome associates with autoimmune diseases and COVID-19. *Nature Genetics* **57,** 65–79 (2025).
12. Kato C, Morimoto S, Takahashi S, **Namba S**, Wang QS, Okada Y & Okano H. Spinal cord motor neuron phenotypes and polygenic risk scores in sporadic amyotrophic lateral sclerosis: deciphering the disease pathology and therapeutic potential of ropinirole hydrochloride. *Journal of Neurology, Neurosurgery & Psychiatry* **96,** 199–201 (2025).
13. Yamamoto K, **Namba S**, Sonehara K, Suzuki K, Sakaue S, Cooke NP *et al.* Genetic legacy of ancient hunter-gatherer Jomon in Japanese populations. *Nature Communications* **15,** 9780 (2024).
14. **Namba S♯**, Akiyama M, Hamanoue H, Kato K, Kawashima M, Kushima I *et al.* Inconsistent embryo selection across polygenic score methods. *Nature Human Behaviour* 2264–2267 (2024).
15. Wang QS, Hasegawa T, Namkoong H, Saiki R, Edahiro R, ..., **Namba S** *et al.* Statistically and functionally fine-mapped blood eQTLs and pQTLs from 1,405 humans reveal distinct regulation patterns and disease relevance. *Nature Genetics* 2054–2067 (2024).
16. Naito T, Inoue K, **Namba S**, Sonehara K, Suzuki K, Matsuda K *et al.* Machine learning reveals heterogeneous associations between environmental factors and cardiometabolic diseases across polygenic risk scores. *Communications Medicine* **4,** 181 (2024).
17. Tomofuji Y, Edahiro R, Sonehara K, Shirai Y, Kock KH, ..., **Namba S** *et al.* Quantification of escape from X chromosome inactivation with single-cell omics data reveals heterogeneity across cell types and tissues. *Cell Genomics* **4,** 100625 (2024).
18. Ojima T, **Namba S**, Suzuki K, Yamamoto K, Sonehara K, Narita A *et al.* Body mass index stratification optimizes polygenic prediction of type 2 diabetes in cross-biobank analyses. *Nature Genetics* **56,** 1100–1109 (2024).
19. De Vincentis A, Tavaglione F, **Namba S**, Kanai M, Okada Y, Kamatani Y *et al.* Poor accuracy and sustainability of the first-step FIB4 EASL pathway for stratifying steatotic liver disease risk in the general population. *Alimentary Pharmacology & Therapeutics* **59,** 1402–1412 (2024).
20. Lo Faro V, Bhattacharya A, Zhou W, Zhou D, Wang Y, ..., **Namba S** *et al.* Novel ancestry-specific primary open-angle glaucoma loci and shared biology with vascular mechanisms and cell proliferation. *Cell Reports Medicine* **5,** 101430 (2024).
21. Suzuki K, Hatzikotoulas K, Southam L, Taylor HJ, Yin X, ..., **Namba S** *et al.* Genetic drivers of heterogeneity in type 2 diabetes pathophysiology. *Nature* **627,** 347–357 (2024).
22. Koyanagi YN, Nakatochi M, **Namba S**, Oze I, Charvat H, Narita A *et al.* Genetic architecture of alcohol consumption identified by a genotype-stratified GWAS and impact on esophageal cancer risk in Japanese people. *Science Advances* **10,** eade2780 (2024).
23. Campos AI, **Namba S**, Lin SC, Nam K, Sidorenko J, Wang H *et al.* Boosting the power of genome-wide association studies within and across ancestries by using polygenic scores. *Nature Genetics* **55,** 1769–1776 (2023).
24. Sato G, Shirai Y, **Namba S**, Edahiro R, Sonehara K, Hata T *et al.* Pan-cancer and cross-population genome-wide association studies dissect shared genetic backgrounds underlying carcinogenesis. *Nature Communications* **14,** 3671 (2023).
25. Tanaka H, Okada Y, Nakayamada S, Miyazaki Y, Sonehara K, **Namba S** *et al.* Extracting immunological and clinical heterogeneity across autoimmune rheumatic diseases by cohort-wide immunophenotyping. *Annals of the Rheumatic Diseases* **83,** 242–252 (2023).
26. Sekita A, Kawasaki H, Fukushima-Nomura A, Yashiro K, Tanese K, ..., **Namba S** *et al.* Multifaceted analysis of cross-tissue transcriptomes reveals phenotype–endotype associations in atopic dermatitis. *Nature Communications* **14,** 6133 (2023).

27. Edahiro R, Shirai Y, Takeshima Y, Sakakibara S, Yamaguchi Y, ..., **Namba S** *et al.* Single-cell analyses and host genetics highlight the role of innate immune cells in COVID-19 severity. *Nature Genetics* **55,** 753–767 (2023).
28. Wang Y, **Namba S**, Lopera E, Kerminen S, Tsuo K, Läll K *et al.* Global Biobank analyses provide lessons for developing polygenic risk scores across diverse cohorts. *Cell Genomics* **3,** 100241 (2023).
29. Tsuo K, Zhou W, Wang Y, Kanai M, **Namba S**, Gupta R *et al.* Multi-ancestry meta-analysis of asthma identifies novel associations and highlights the value of increased power and diversity. *Cell Genomics* **2,** 100212 (2022).
30. **Namba S***, Saito Y*, Kogure Y, Masuda T, Bondy ML, Gharahkhani P *et al.* Common germline risk variants impact somatic alterations and clinical features across cancers. *Cancer Research* **83,** 20–27 (2022).
31. Zhou W, Kanai M, Wu KHH, Rasheed H, Tsuo K, ..., **Namba S** *et al.* Global Biobank Meta-analysis Initiative: Powering genetic discovery across human disease. *Cell Genomics* **2,** 100192 (2022).
32. **Namba S***, Konuma T*, Wu KH, Zhou W & Okada Y. A practical guideline of genomics-driven drug discovery in the era of global biobank meta-analysis. *Cell Genomics* **2,** 100190 (2022).
33. Mishra A*, Malik R*, Hachiya T*, Jürgenson T*, **Namba S***, Posner DC* *et al.* Stroke genetics informs drug discovery and risk prediction across ancestries. *Nature* **611,** 115–123 (2022).
34. Yamamoto K, Sonehara K, **Namba S**, Konuma T, Masuko H, Miyawaki S *et al.* Genetic footprints of assortative mating in the Japanese population. *Nature human behaviour* **7,** 65–73 (2022).
35. Wang QS, Edahiro R, Namkoong H, Hasegawa T, Shirai Y, ..., **Namba S** *et al.* The whole blood transcriptional regulation landscape in 465 COVID-19 infected samples from Japan COVID-19 Task Force. *Nature communications* **13,** 4830 (2022).
36. Namkoong H, Edahiro R, Takano T, Nishihara H, Shirai Y, ..., **Namba S** *et al.* DOCK2 is involved in the host genetics and biology of severe COVID-19. *Nature* **609,** 754–760 (2022).
37. Shirai Y, Nakanishi Y, Suzuki A, Konaka H, Nishikawa R, ..., **Namba S** *et al.* Multi-trait and cross-population genome-wide association studies across autoimmune and allergic diseases identify shared and distinct genetic component. *Annals of the rheumatic diseases* **81,** 1301–1312 (2022).
38. Ho WK, Tai MC, Dennis J, Shu X, Li J, ..., **Namba S** *et al.* Polygenic risk scores for prediction of breast cancer risk in Asian populations. *Genetics in medicine* **24,** 586–600 (2022).
39. **Namba S**, Ueno T, Kojima S, Kobayashi K, Kawase K, Tanaka Y *et al.* Transcript-targeted analysis reveals isoform alterations and double-hop fusions in breast cancer. *Communications biology* **4,** 1320 (2021).
40. **Namba S**, Sato K, Kojima S, Ueno T, Yamamoto Y, Tanaka Y *et al.* Differential regulation of CpG island methylation within divergent and unidirectional promoters in colorectal cancer. *Cancer science* **110,** 1096–1104 (2019).

総説（日本語）

1. **Namba S** & Okada Y. [CURRENT PIPELINES FOR WHOLE-GENOME SEQUENCING ANALYSES]. *Arerugi* **72,** 1110–1112 (2023).
2. **Namba S** & Okada Y. [UTILIZING GENOMIC INFORMATION FOR BIOMARKER IDENTIFICATION AND GENOMIC DRUG DISCOVERY]. *Arerugi* **69,** 952–957 (2020).